BS25102-Engineering Mathematics-II

(Ordinary Differential Equations, Integral Calculus and Statistics)

Assignment on Unit-1: First Order Ordinary Differential Equations

Write the complete Summary of Unit-1 [First Order Ordinary Differential Equations].

Type A: Exact Differential Equations

1. Solve $(2xy^4 + siny)dx + (4x^2y^3 + xcosy)dy = 0$.

   [**Answer:** $x^2y^4 + xsiny = c$]

2. Solve $(3y^2 + 10xy^2)dx + (6xy - 2 + 10x^2y)dy = 0$.

   [**Answer:** $3xy^2 + 5x^2y^2 - 2y = c$]

3. Solve $(ysec^2x + secx.tanx)dx + (tanx + 2y)dy = 0$.

   [**Answer:** $y.\tan x + secx + y^2 = c$]

Type B: Reducible to Exact Differential Equations

4. Solve $(x^2y - 2xy^2)dx + (3x^2y - x^3)dy = 0$.

   [**Answer:** $\frac{x}{y} + \log\left(\frac{y^3}{x^2}\right) = c$]

5. Solve $(xy + y^2)dx - x^2dy = 0$.

   [**Answer::** $\frac{x}{y} + \log x = c$]

6. Solve $(x^2y^2 + 5xy + 2)ydx + (x^2y^2 + 4xy + 2)xdy = 0$.

   [**Answer:** $xy + \log\left(\frac{x^5}{y^4}\right) - \frac{2}{xy} = c$]

7. Solve $(x^2y^2 + 2)ydx + (2 - 2x^2y^2)xdy = 0$.

   [**Answer:** $\log\left(\frac{x}{y^2}\right) - \frac{1}{x^2y^2} = c$]

8. Solve $(x^2 + y^2 + 1)dx - 2xydy = 0$.

   [**Answer:** $x^2 - y^2 - 1 = cx$]

9. Solve $(x^2 + y^2 + x)dx + xydy = 0$.

   [**Answer:** $3x^4 + 6x^2y^2 + 4x^3 = c_1$]

10. Solve $(2x^2y^2 + ye^x)dx + (-e^x - y^3)dy = 0$.

    [**Answer:** $\frac{2x^3}{3} + \frac{e^x}{y} - \frac{y^2}{2} = c$]

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11. Solve $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$.

[**Answer:** $x\left(y + \frac{2}{y^2}\right) + y^2 = c$]

12. Solve $(y^3 - 2x^2y)dx + (2xy^2 - x^3)dy = 0$.

[**Answer:** $x^2y^4 - x^4y^2 = c_1$]

Type C: Linear Differential Equations

13. Solve $x^2 \frac{dy}{dx} = 3x^2 - 2xy + 1$.

[**Answer:** $yx^2 = x^3 + x + c$]

14. Solve $(x^2 + 1)\frac{dy}{dx} + 4xy = \frac{1}{(x^2+1)^2}$.

[**Answer:** $y(x^2 + 1)^2 = tan^{-1}x + c$]

15. Solve $\frac{dy}{dx} = \frac{e^x - 3xy}{x^2}$.

[**Answer:** $x^3y = (x - 1)e^x + c$]

16. Solve $xcosx\frac{dy}{dx} + (cosx - xsinx)y = 1$.

[**Answer:** $yxcosx = x + c$]

Type D: Reducible to Linear Differential Equations

17. Solve $xy - \frac{dy}{dx} = y^3e^{-x^2}$.

[**Answer:** $\frac{e^{x^2}}{y^2} = 2x + c$]

18. Solve $\frac{dy}{dx} - \frac{tany}{1+x} = (1 + x)e^x secy$.

[**Answer:** $\frac{siny}{1+x} = e^x + c$]

19. Solve $cosy - x\ siny = sec^2x$.

[**Answer:** $xcosy = tanx + c$]

20. Solve $3y^2\frac{dy}{dx} + 2xy^3 = 4xe^{-x^2}$.

[**Answer:** $y^3e^{x^2} = 2x^2 + c$]

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